Unified Engineering Specification: AI & Cyber Phishing Detection

Integrated Methodology • Two-Tier Workflow • 4-Phase End-to-End System Architecture

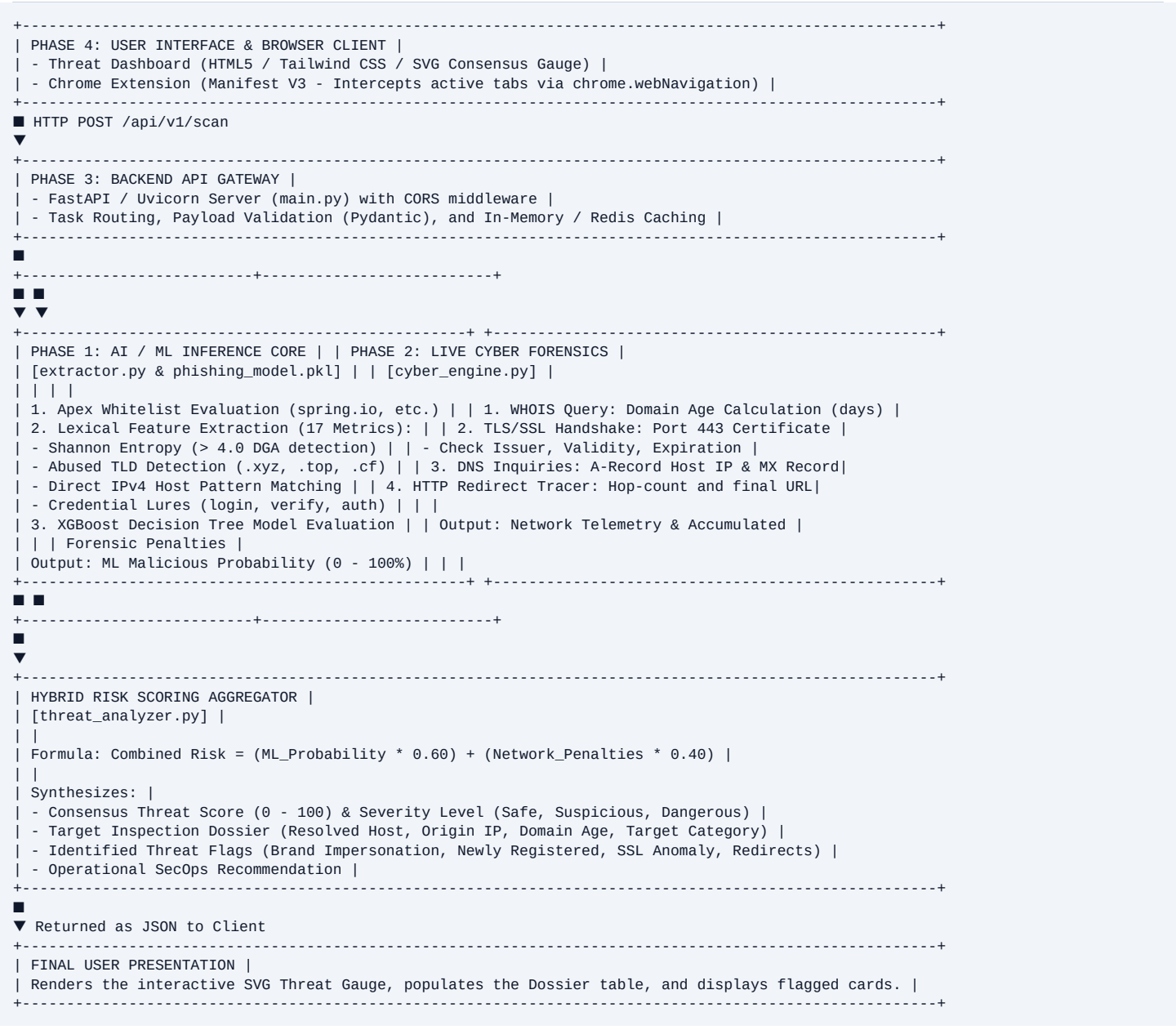
1. Combined Methodology (Phase 1 AI + Phase 2 Cyber Forensics)

Modern phishing attacks deploy fast-flux DNS and short-lived domains that render static blacklists obsolete. Our defense uses a **hybrid layered methodology**: combining zero-network lexical feature analysis with active network-layer digital forensics to catch deceptive zero-day attacks while suppressing false alarms.

Layer	Methodology & Principles	Key Security Mechanics
Tier 0: Authority Whitelist	Deterministic short-circuiting to bypass statistical anomalies on verified global ecosystem authorities	Root apex verification against trusted registries (sprin
Tier 1: Lexical ML Engine (Phase 1)	Probabilistic classification of raw URL structures without network latency.	17 topological features (i.e. Shannon entropy, abused TL
Tier 2: Cyber Forensics (Phase 2)	Active digital investigation of infrastructure to evaluate domain legitimacy	WHOIS domain creation age, TLS/SSL socket hands
Tier 3: Hybrid Aggregator	Mathematical scoring combining machine learning confidence and network penalties	Final Risk = (ML_Prob * 0.60) +

2. Integrated Operational Workflow (Phase 1 to Phase 2)

Step	Engine Phase	Action / Execution Step	Produced Output / Telemetry
1. Intake	Ingestion	Receive raw URL target via API or User Dashboard.	Sanitized URL string (with protocol scheme
2. Filter	Tier 0 Check	Extract registered apex domain and verify against trusted authority 'SAFE'	Immediate Safety Verdict (0.01%)' verdict if matche
3. Vectorize	Phase 1 (AI)	Parse string into 17 numerical values (entropy, TLD abuse, character ratios).	Aligned 1x17 Feature Matrix
4. Predict	Phase 1 (AI)	Pass vector through trained serialized XGBoost model.	ML Malicious Probability (0% - 100%)
5. Probe	Phase 2 (Cyber)	Query WHOIS registry for creation timestamp and registration age.	Domain Age in Days (flags age < 30 days)
6. Handshake	Phase 2 (Cyber)	Open TLS connection on port 443; extract X.509 certificate headers.	SSL Issuer Identity & Expiry Status
7. Resolve	Phase 2 (Cyber)	Resolve DNS A-records for Host IP and check for active MX mail records.	Mail Server Profile
8. Trace	Phase 2 (Cyber)	Follow HTTP 301/302 redirects with custom browser User-Agent header.	Final Destination URL
9. Aggregate	Hybrid Fusion	Fuse ML probability (60%) with accumulated network penalties (40%).	Combined Threat Index (Score 0 - 100)
10. Render	Phase 3/4 (UI)	Map risk score, dossier, and threat flags to the frontend SOC interface.	Dynamic Dashboard & Threat Cards

3. Complete Structural Architecture Diagram (All 4 Phases)

System connectivity between Frontend, Backend Gateway, Cyber Forensics, and AI Model